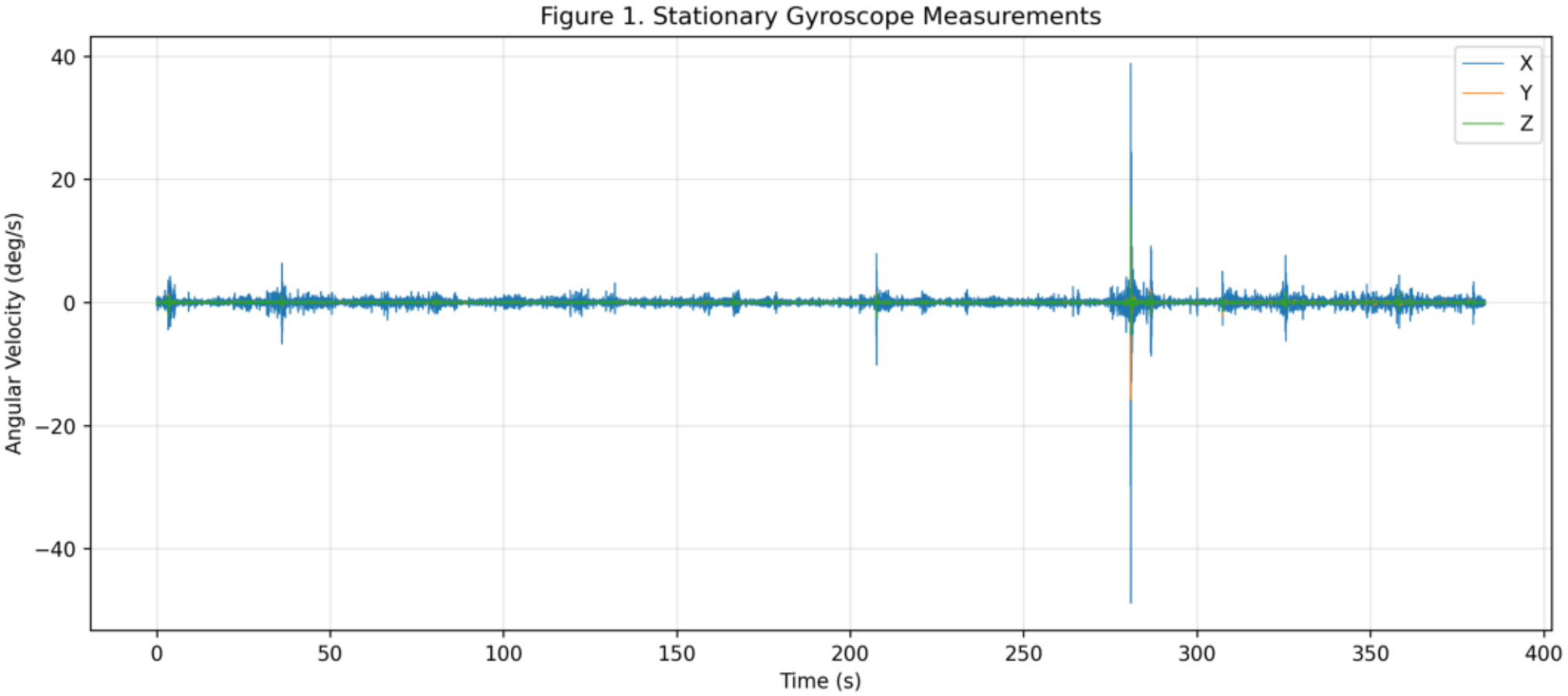
EECE 5554 — Lab 3

VN-100 IMU Analysis

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github.com/greengreerfine/EECE5554


Figure 1. Stationary Gyroscope Measurements
Angular Velocity (deg/s)
Time (s)
X
Y
Z
40
20
0
−20
−40
0
50
100
150
200
250
300
350
400

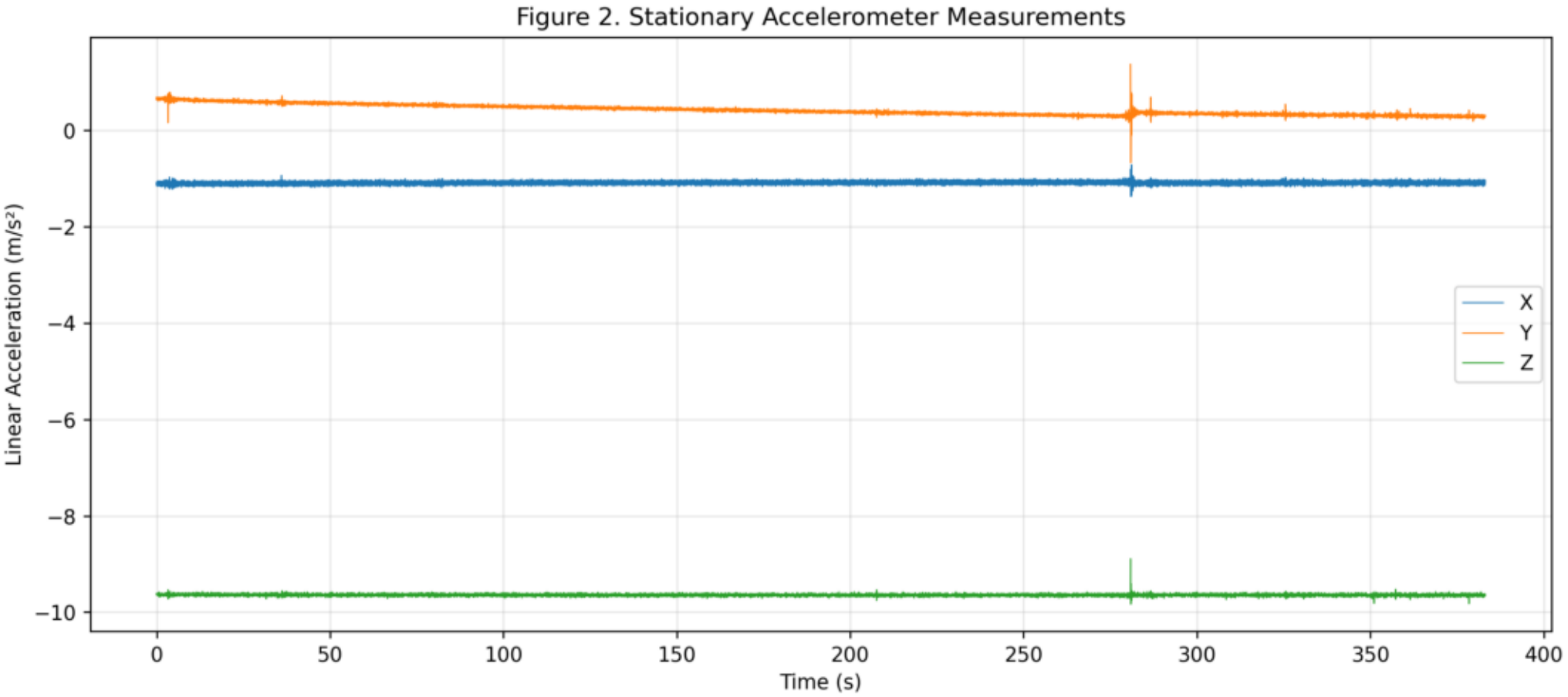

Figure 2. Stationary Accelerometer Measurements
Linear Acceleration (m/s²)
0
−2
−4
−6
−8
−10
0
50
100
150
200
250
300
350
400
Time (s)
X
Y
Z

Figure 3. Stationary Orientation

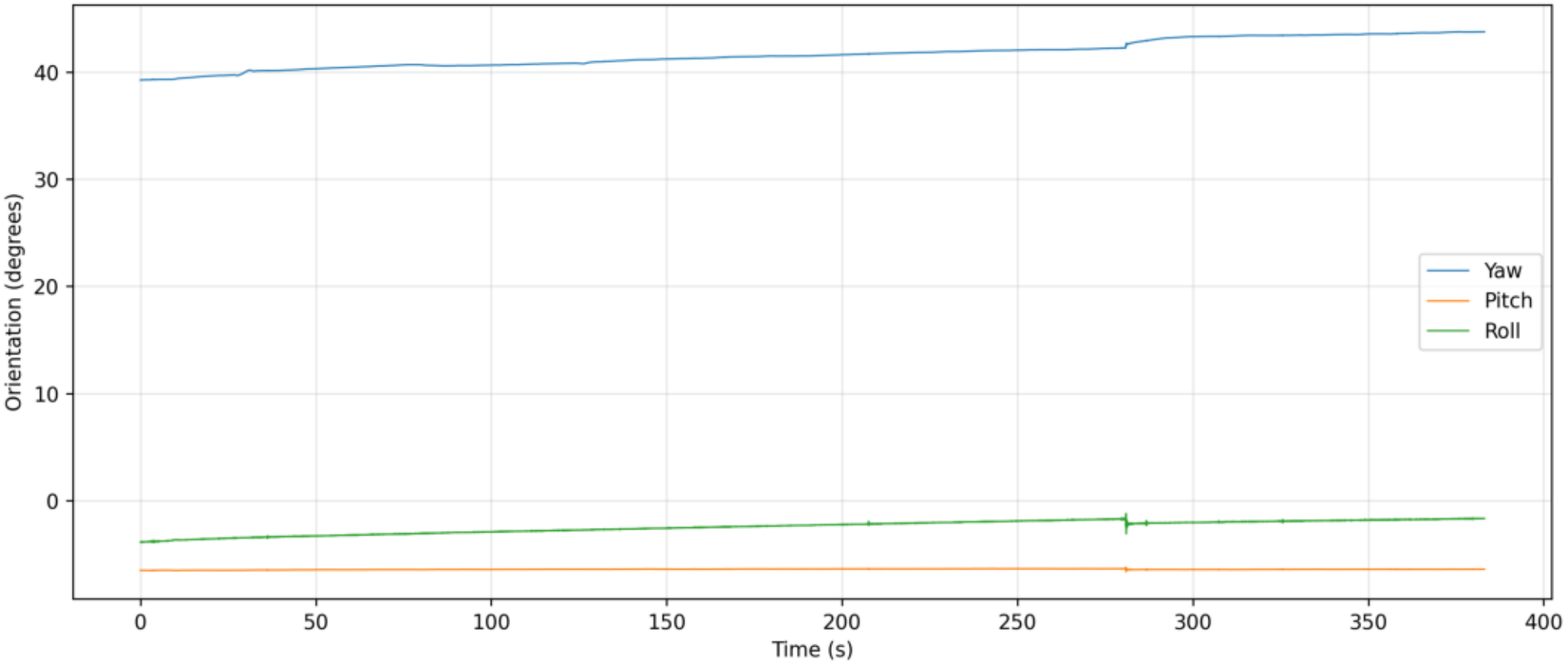


Figure 4. Stationary Sensor Noise Distributions

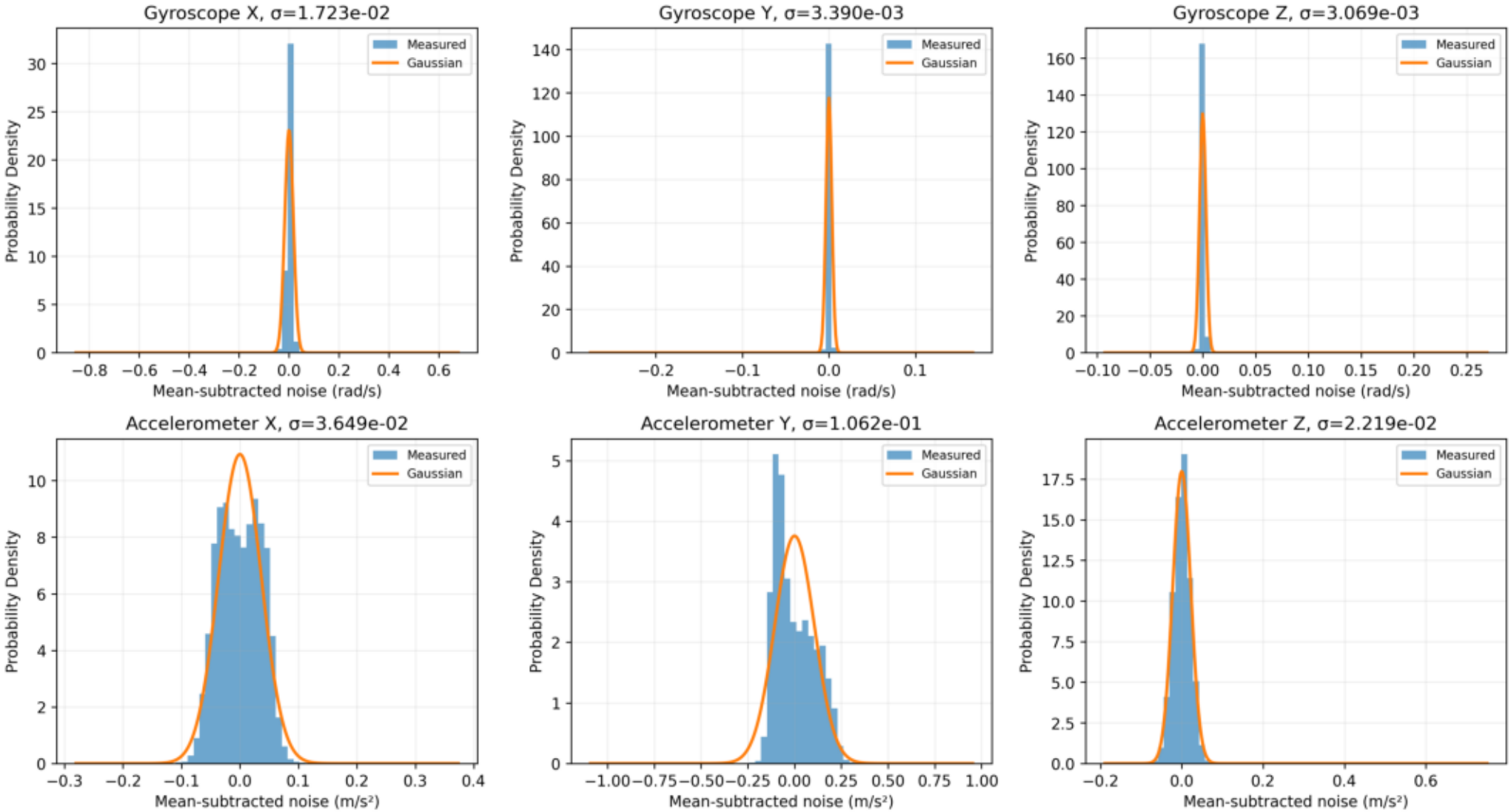


Figure 5. Allan Deviation of VN-100 Gyroscope

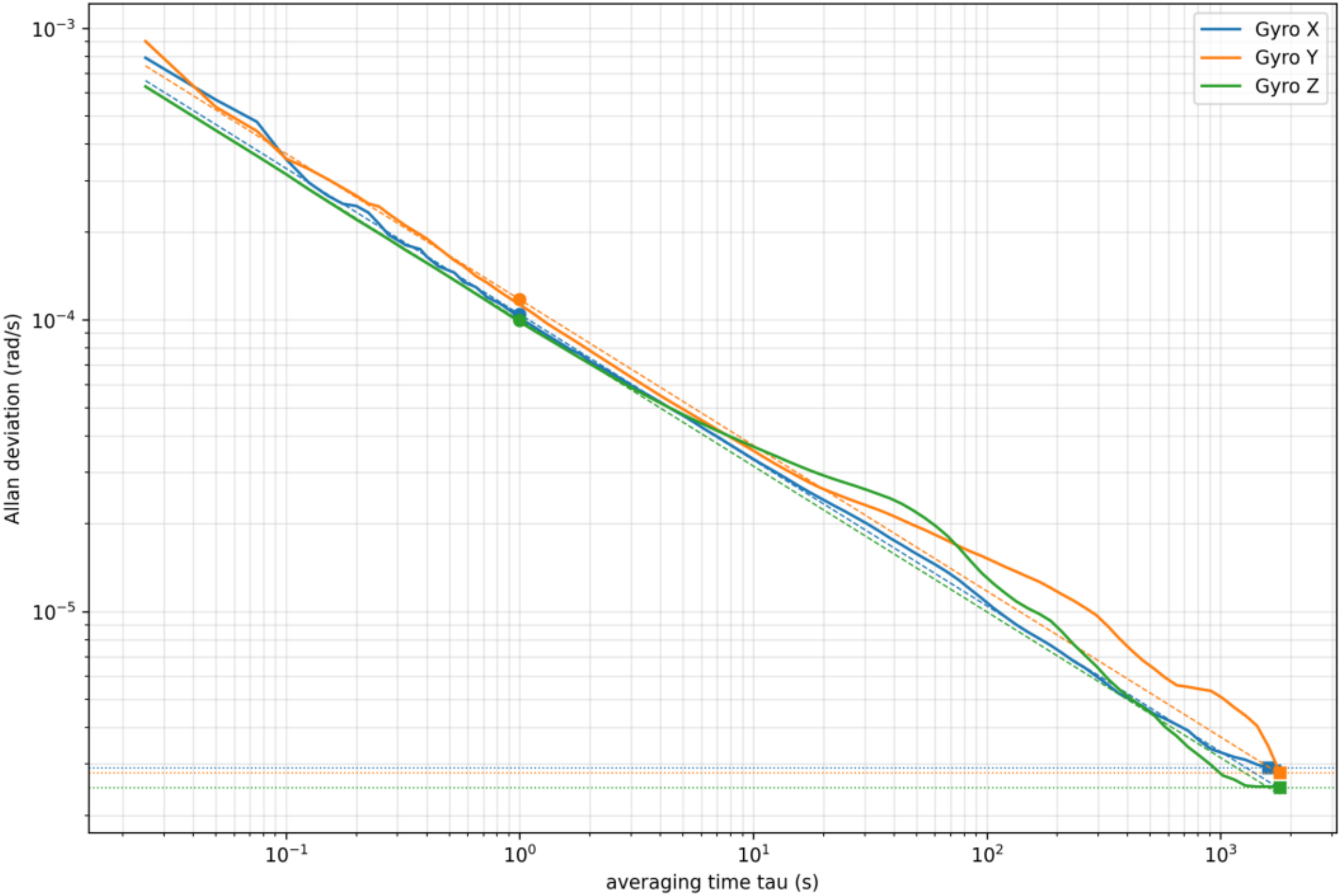


Figure 6. VN-100 Motion Experiment

Figure 6a. Horizontal Turn

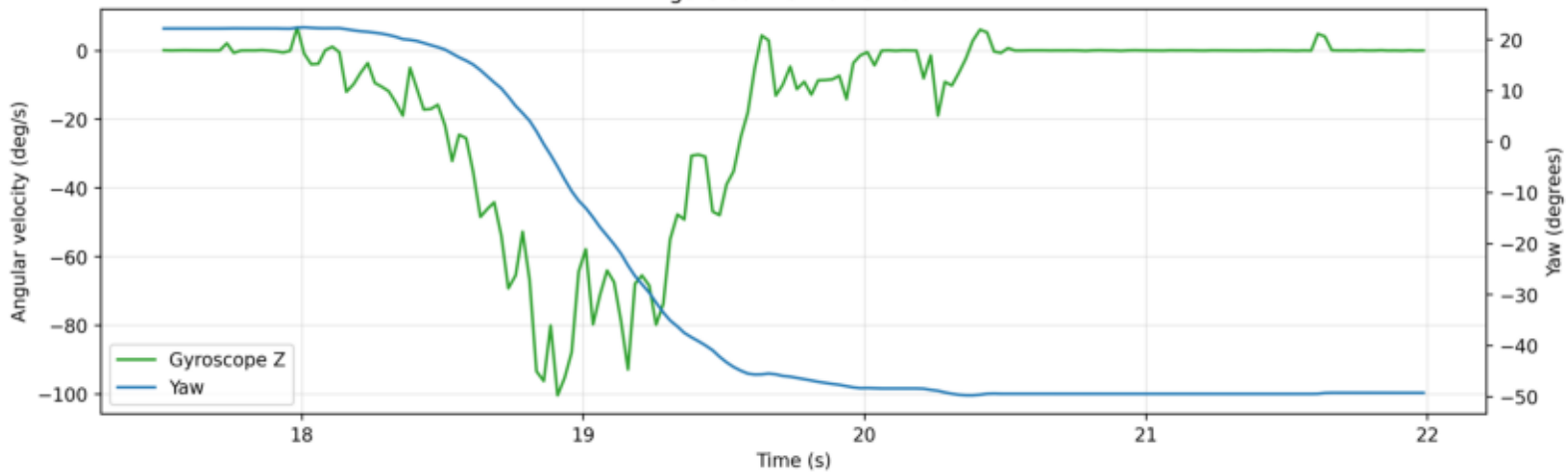


Figure 6b. Tip Onto Side

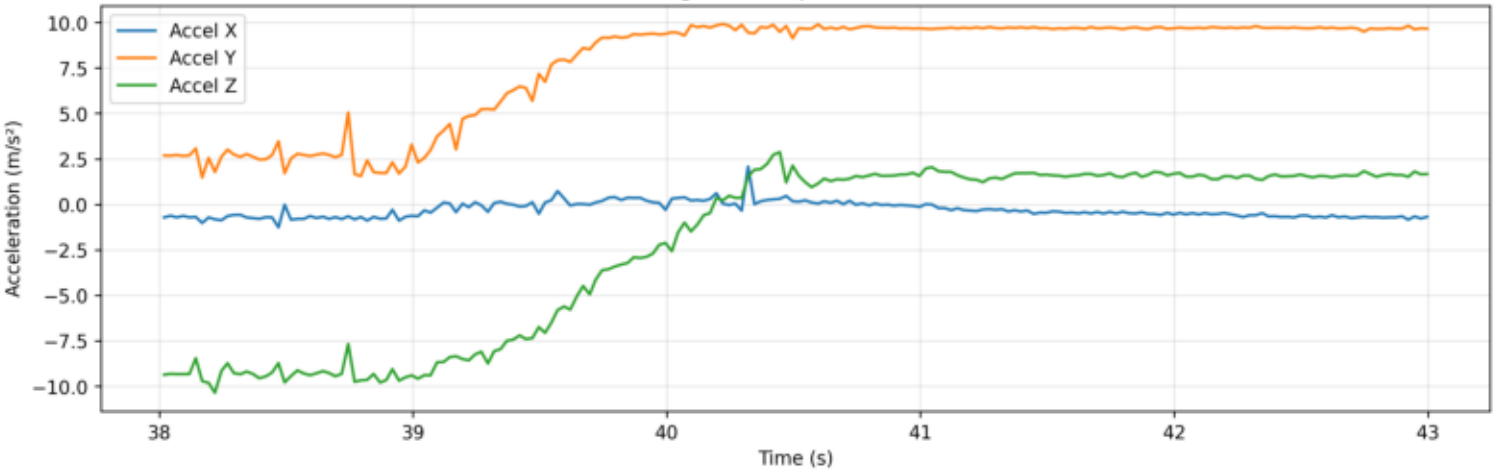


Figure 6c. Table Taps: Acceleration Transients

